INTERNSHIP PROJECT REPORT • B.TECH CSE, IILM UNIVERSITY

Development of an AI-Powered Client Query Automation & Document Classification Tool

A proof-of-concept NLP system built during a Summer Internship at J. D. Jones & Co. (P) Ltd., Kolkata

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CHAPTER 4.1 – 4.2

Organisation Profile

J. D. Jones & Co. (P) Ltd. is an ISO 9001:2015 certified company based in Kolkata, engaged in shipping and allied trading services. Its AI Development team builds small-scale automation solutions to support day-to-day operations.

Shipping & Trading | ISO 9001:2015 | AI Development Team

Company

J. D. Jones & Co. (P) Ltd.

CIN: U10200WB1923PTC004802

At a Glance

COMPANY

J. D. Jones & Co. (P) Ltd.

INDUSTRY

Shipping, Trading & Allied Business Services

REGISTERED OFFICE

Kolkata, West Bengal, India

CERTIFICATION

ISO 9001:2015 (DNV·GL)

ROLE

Intern – AI Developer

DURATION

6 Aug 2026 – 5 Nov 2026 (3 months)

CHAPTER 4.3

Problem Statement

“The operations team receives a large number of routine client queries daily — on shipment status, documentation, and billing — answered manually, which is time-consuming and repetitive.”

1

Manual & Repetitive

Staff read and answer recurring query categories by hand every day.

2

Time-Consuming

Predictable, high-volume categories consume disproportionate staff time.

3

Human-in-the-Loop Needed

Any automation must keep a human reviewer before replies reach clients.

CHAPTER 4.4

Project Objectives

1. Gain hands-on, industry-level experience applying AI/NLP to a real business problem
2. Design and develop a prototype that classifies incoming client queries by category
3. Build a module that suggests draft responses for common, repetitive queries
4. Develop a basic document-tagging utility to organise incoming business documents
5. Understand professional development practices, version control, and team collaboration

CHAPTER 4.5

Scope of the Project

The project was scoped as an internal prototype for the operations team — built and tested on a sample, anonymised dataset rather than deployed to production.

IN SCOPE

- ✓ Classification prototype on sample data
- ✓ Measuring accuracy & F1-score
- ✓ Response-suggestion drafting aid
- ✓ Document-tagging utility
- ✓ Time-saving estimation on repetitive tasks

OUT OF SCOPE

- ✗ Production deployment
- ✗ Live client communication integration
- ✗ Fully automated reply sending

CHAPTER 4.6

Technologies & Tools Used

LANGUAGE

Python 3

NLP

NLTK / spaCy, scikit-learn

DATA HANDLING

Pandas, NumPy

MODELLING

TF-IDF + Logistic Regression

INTERFACE

Flask (internal dashboard)

VERSIONING

Git & GitHub

DEV ENVIRONMENT

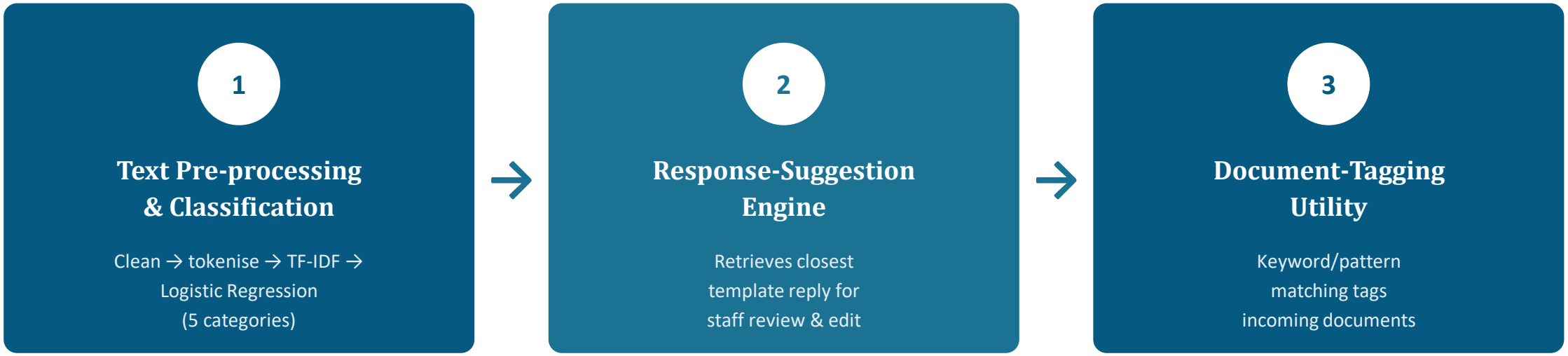
VS Code, Jupyter Notebook

TRACKING

Excel, Google Sheets

CHAPTER 4.7

System Architecture



All three modules are exposed through a lightweight internal Flask dashboard — a team member pastes a query or document snippet and instantly sees the predicted category, confidence, and a suggested draft response.

CHAPTER 4.8

Methodology

1 Requirement Understanding

Shadowed the operations team; identified frequent, time-consuming query categories.

2 Data Collection & Preparation

Compiled a sample, anonymised dataset tagged across 5 categories; tokenised, stop-words removed, TF-IDF applied.

3 Model Development

Compared Logistic Regression, Naive Bayes and SVM on a held-out split; TF-IDF + Logistic Regression selected.

4 Response-Suggestion Module

Rule-assisted module retrieves the closest template reply for staff review — not fully automated.

5 Document Tagging Utility

Keyword/pattern matching automatically tags incoming documents to speed up sorting.

6 Testing & Internal Demo

Wrapped in a Flask dashboard for team testing; feedback incorporated in iterative updates.

CHAPTER 4.8.2

Sample Tagged Dataset

Client queries were tagged across five recurring categories to train and test the classifier.

Query Text	Category
Please share the latest location of cargo JDJ2584.	Shipment Status
Can you clarify the additional charges on bill JDJ7252?	Billing Query
The documents for JDJ4510 were sent with incorrect details.	Complaint / Escalation
Please forward the insurance certificate for order JDJ4483.	Documentation Request
Who should I contact regarding new business enquiries?	General Enquiry

CHAPTER 4.8.6

Model Performance

93.8%
Overall Accuracy

0.94
Weighted F1-Score

40–50%
Faster Drafting (est.)

Per-Category Precision / Recall / F1

Category	Precision	Recall	F1
Billing Query	1.00	1.00	1.00
Complaint / Escalation	0.83	1.00	0.91
Documentation Request	0.87	0.93	0.90
General Enquiry	1.00	1.00	1.00
Shipment Status	1.00	0.81	0.90

Key Observation

Most confusion occurred between “Shipment Status” and “Documentation Request”, since delay queries often also referenced missing documents.

CHAPTER 4.9

Expected Outcomes

1

Working Prototype

Classifies client queries into the correct category with ~94% test accuracy.

2

Reduced Manual Effort

Faster drafting of responses for common, repetitive client queries.

3

Document Sorting Aid

A basic tagging utility speeds up organisation of incoming documents.

4

Feasibility Demonstrated

Clear case for expanding into production, with a larger training dataset.

5

Personal Growth

Strengthened Python, NLP, scikit-learn, Flask, time management & teamwork skills.

Thank You

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